Lab - 9.3.8 - Exploring Nmap

Ahmed ABDELHEDI

Part 1: Exploring Nmap

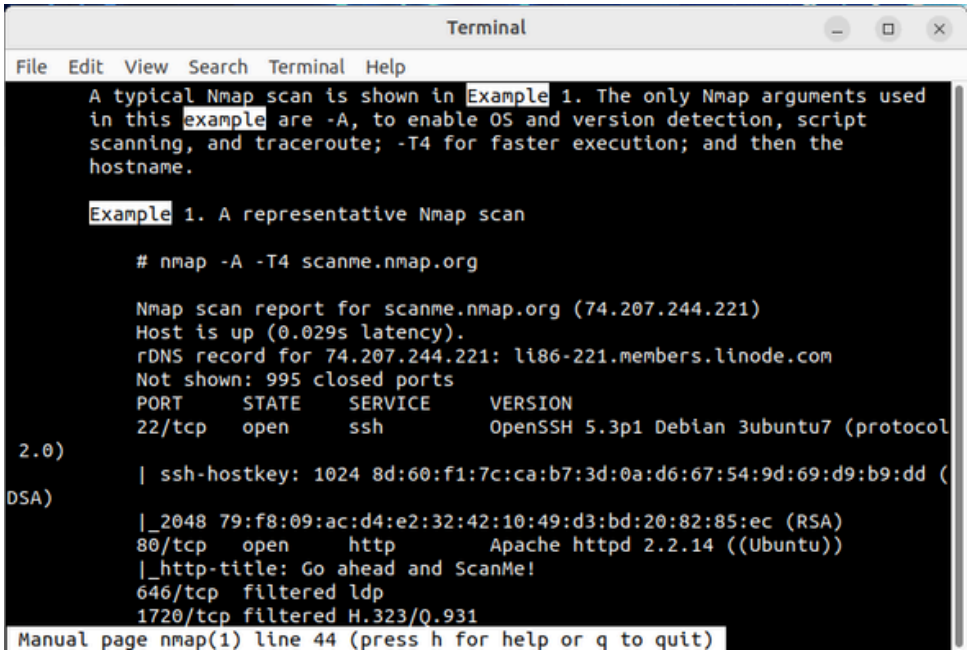
What is Nmap?

Nmap (“Network Mapper”) is an open source tool for network exploration and security auditing

What is nmap used for?

It was designed to rapidly scan large networks,
although it works fine against single hosts. Nmap uses raw IP packets in novel ways to determine what hosts are available on the network, what services (application name and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters/firewalls are in use, and dozens of other characteristics.

d+e:



What is the nmap command used?

nmap -A -T4 scanme.nmap.org

What does the switch -A do?

to enable OS and version detection, script scanning, and traceroute

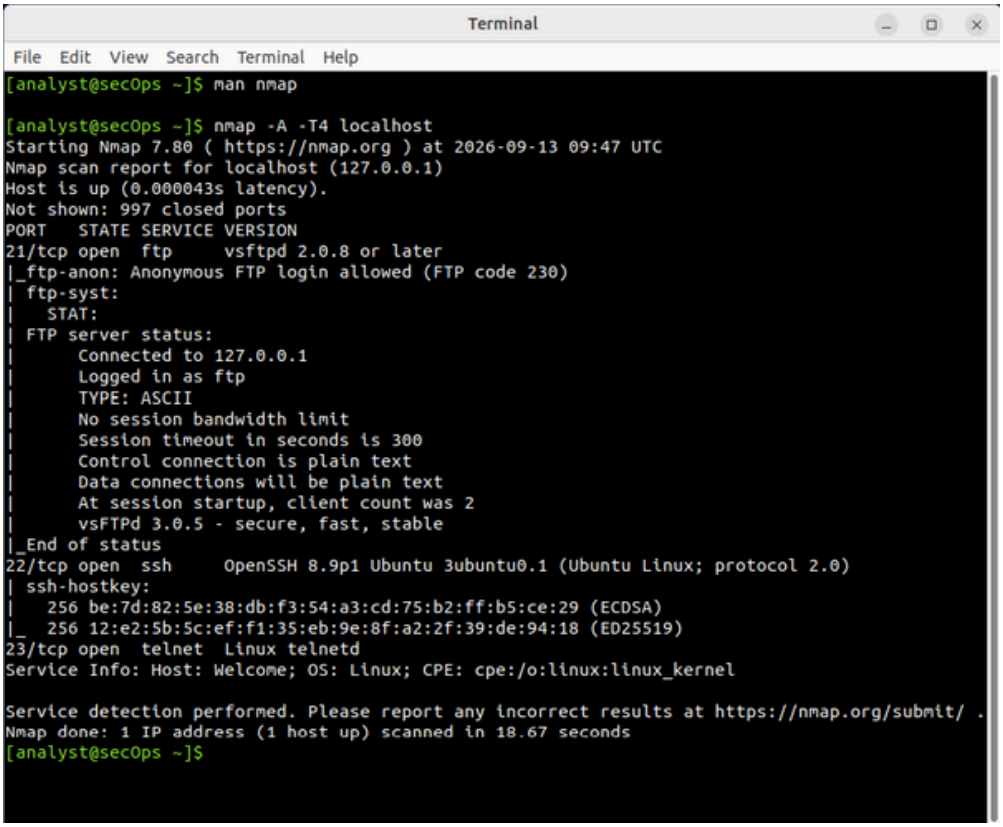
What does the switch -T4 do?

for faster execution

Part 2: Scanning for Open Ports

Step 1: Scan your localhost.

a:



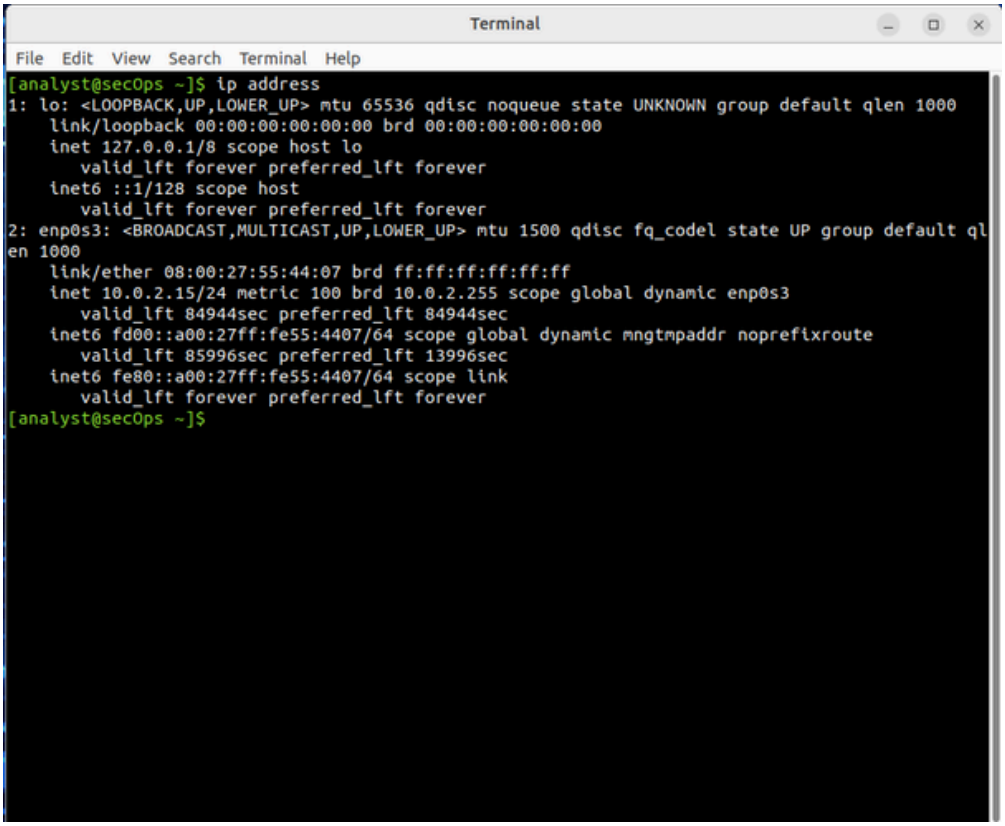
Which ports and services are opened?

For each of the open ports, record the software that is providing the services

Port	Service	Software providing the service
21/tcp	FTP	**vsftpd 3.0.5**
22/tcp	SSH	**OpenSSH 8.9p1 Ubuntu 3ubuntu0.1**
23/tcp	Telnet	**Linux telnetd**

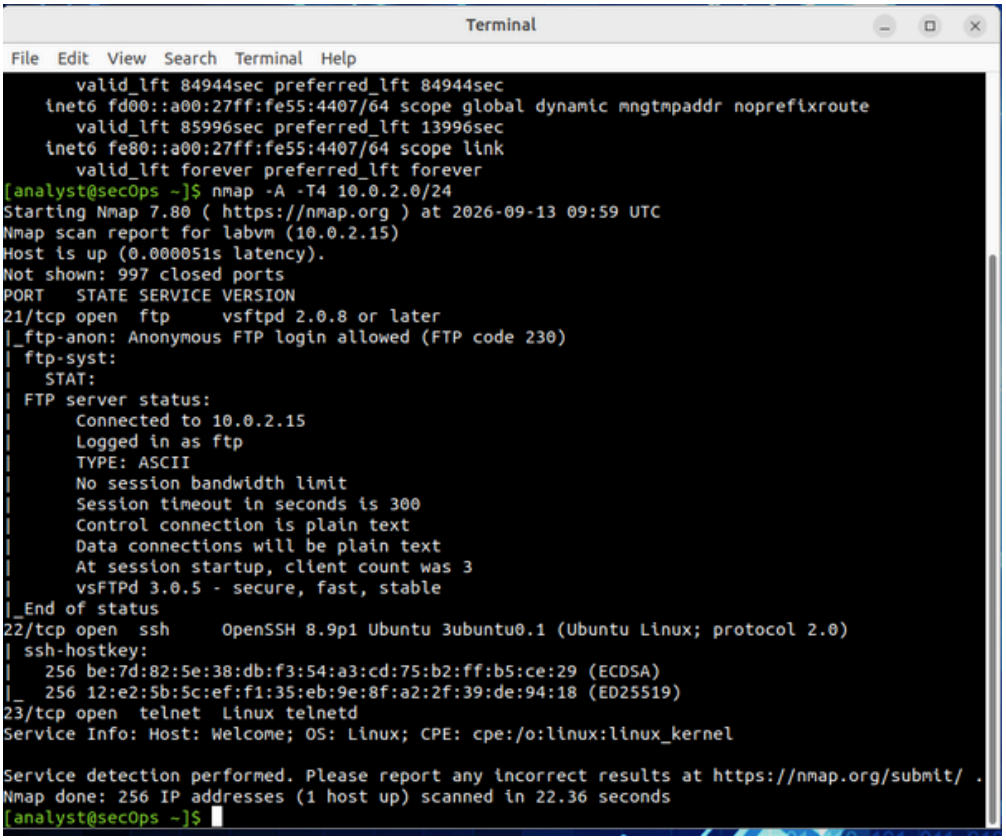
Step 2: Scan your network.

a:



VM IP address: 10.0.2.15
Subnet mask: 255.255.255.0 (/24)
Network: 10.0.2.0/24

b:



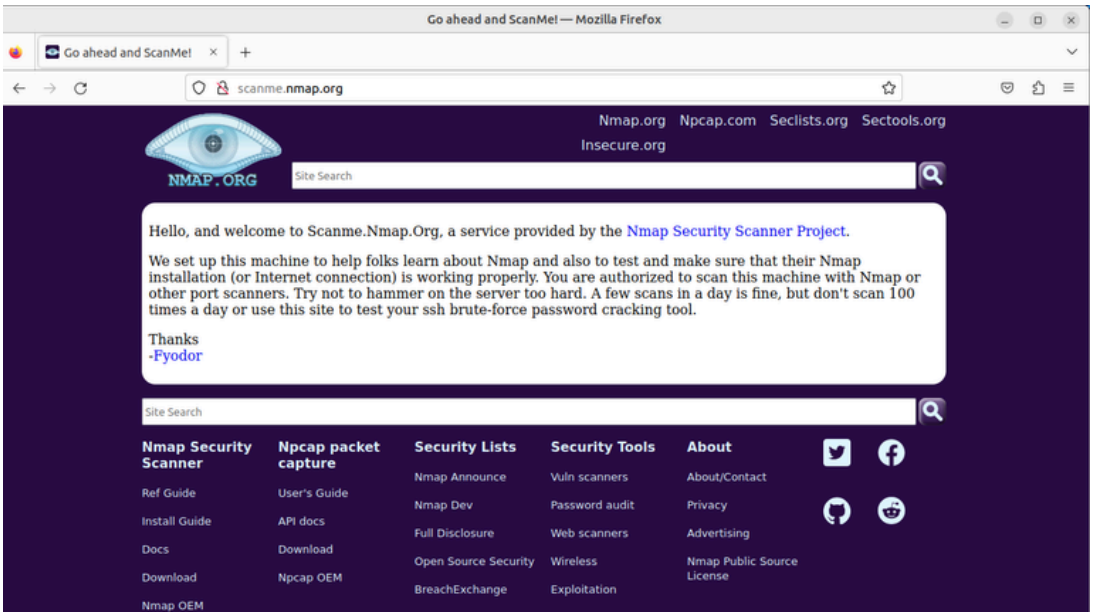
How many hosts are up?

From your Nmap results, list the IP addresses of the hosts that are on the same LAN as your VM. List some of the services that are available on the detected hosts.

- **Hosts up: 1**
- **Host_ip: 10.0.2.15**
- **Services: FTP, SSH, and Telnet**

Step 3: Scan a remote server.

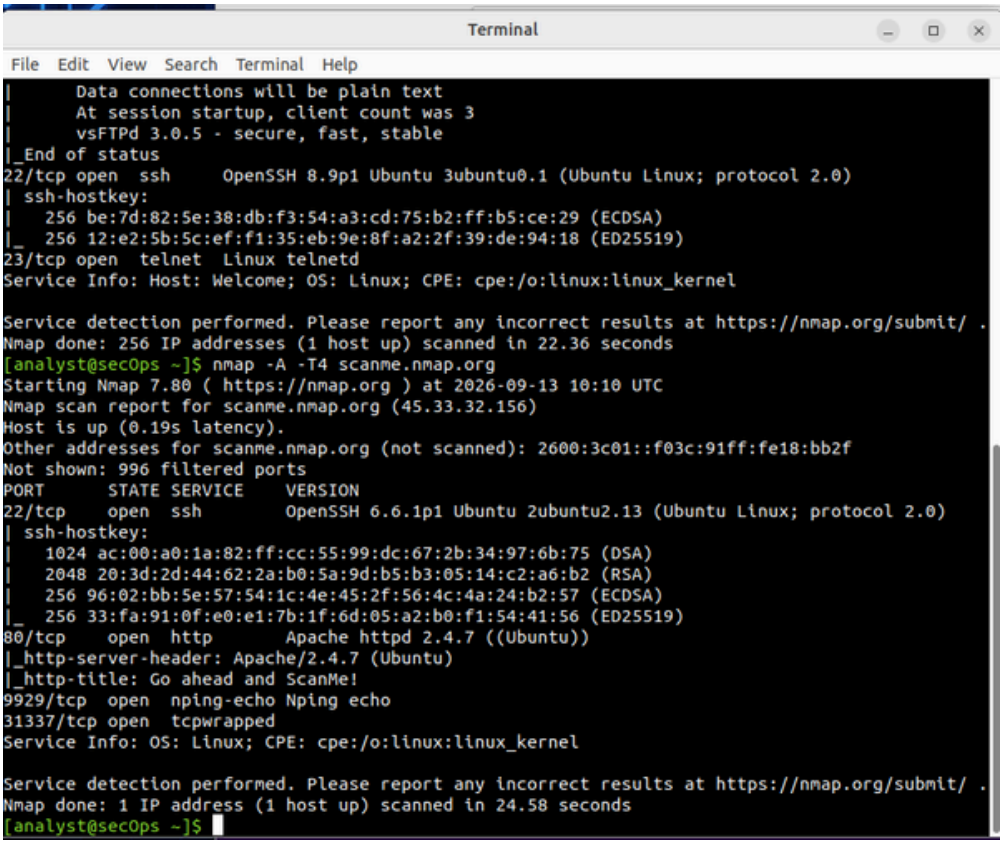
a:



What is the purpose of this site?

Learn how to use Nmap.
Test whether their Nmap installation and internet connection work correctly.
Practise authorized port scanning on a safe, public test server.

b:



Which ports and services are opened?

22/tcp open ssh

80/tcp open http

9929/tcp open nping-echo

31337/tcp open tcpwrapped

Which ports and services are filtered?

996 ports are filtered, but Nmap does not list their individual port numbers or services

What is the IP address of the server?

45.33.32.156

What is the operating system?

Linux, specifically Ubuntu

Reflection Question:
Nmap is a powerful tool for network exploration and management. How can Nmap help with network security? How can Nmap be used by a threat actor as a nefarious tool?

Nmap helps defenders discover devices, open ports, running services, and security weaknesses. Threat actors can use the same information to identify vulnerable systems and plan attacks.